HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 04 Name: _____ Marks: / 42

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If a, b, c are non-zero constants and $x(x+3a)+a \equiv x^2+2(bx+c)$, then $a : b : c =$

   A. $2 : 3 : 1$

   B. $2 : 3 : 4$

   C. $3 : 2 : 6$

   D. $6 : 4 : 3$

(DSE-2013 PII Q08)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. Susan sells two cars for $80 080 each. She gains 30% on one and loses 30% on the other. After the transactions, Susan

   A. loses $15 840.

   B. gains $5 544.

   C. gains $10 296.

   D. has no gain and no loss.

(DSE-2013 PII Q10)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. The actual area of a playground is $900\,m^2$. Its area on a map is $36\,cm^2$. Then the scale of the map is

A. 1 : 25

B. 1 : 50

C. 1 : 500

D. 1 : 250 000

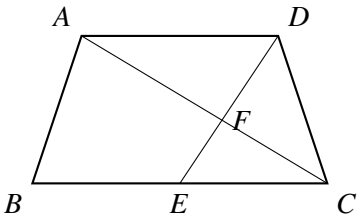
(DSE-2013 PII Q12)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. In the figure, $ABCD$ is a trapezium with $AD//BC$, and $AD : BC = 2 : 3$. Let E be the midpoint of BC. AC and DE intersect at F. If the area of ΔCEF is $36\,\text{cm}^2$, then the area of the trapezium $ABCD$ is.



A. $216\,\text{cm}^2$

B. $264\,\text{cm}^2$

C. $280\,\text{cm}^2$

D. $320\,\text{cm}^2$

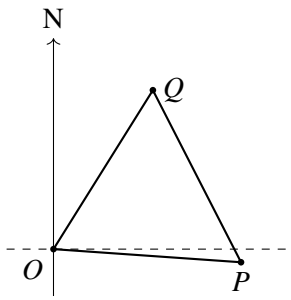
(DSE-2013 PII Q18)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. In the figure, the bearing of P from O is $S86°E$ and the bearing of from O is $N32°E$. If P and Q are equidistant from O, then the bearing of P from Q is



A. N24°W

B. N27°W

C. S24°E

D. S27°E

(DSE-2013 PII Q20)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. If $0^\circ < x < 90^\circ$, which must be true?

I. $\tan x \tan(90^\circ - x) = 1$

II. $\sin x - \sin(90^\circ - x) < 0$

III. $\cos x + \cos(90^\circ - x) > 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2013 PII Q23)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. The coordinates of the points A and B are $(2,5)$ and $(4,-1)$ respectively. Let P be a moving point in the rectangular coordinate plane such that $AP = BP$. Find the equation of the locus of P.

A. $x-3y+3=0$

B. $x-3y-7=0$

C. $x-3y+13=0$

D. $3x+y-11=0$

(DSE-2013 PII Q24)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. The equation of the circle C is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points P and Q are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of C is 5.

II. The midpoint of PQ lies outside C.

III. If G is the centre of C, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. If the mean and mode of nine numbers $14, 6, 4, 5, 7, 5, x, y, z$ are 8 and 14 respectively, then the median of these nine number is

A. 5.

B. 6.

C. 7.

D. 8.

(DSE-2013 PII Q27)

Answers written in the margins will not be marked.

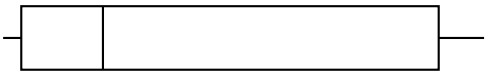
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

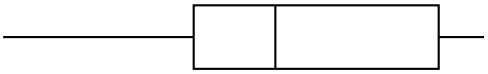
10. The stem-and-leaf diagram shows hourly wages. Which box-and-whisker diagram may represent them?

4	0 2 2 2 4 4 4 7
5	0 0 1 2 2 6 8 9
6	3 5 5 7
7	0
8	2 6
9	5

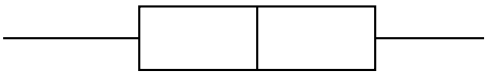
A.



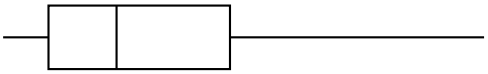
B.



C.



D.



(DSE-2013 PII Q29)

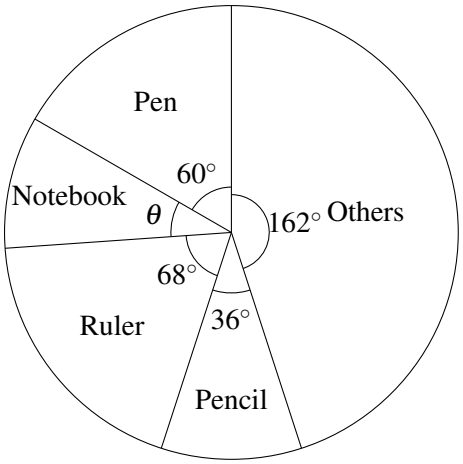
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

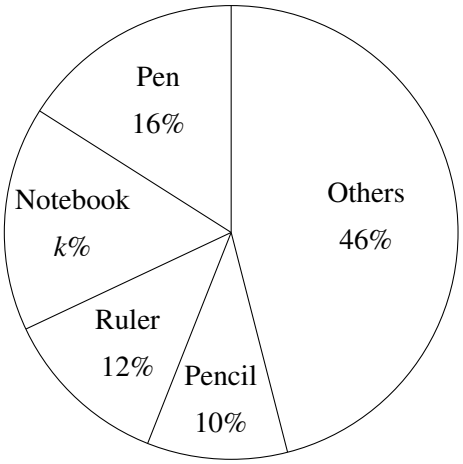
Answers written in the margins will not be marked.

11. The pie charts below show the distributions of the profits of stationery shop$X$ and stationery shop Y from the sales of stationery in a certain month. Which of the following must be true?

Distribution of the profits of stationery shop X



Distribution of the profits of stationery shop Y



A. In that month, the profit from the sales of pencils of stationery shop X is the same as that of stationery shop Y.

B. In that month, the total profit from the sales of pens and notebooks of stationery shop X is less than the total profit from the sales of rulers and pencils of the shop.

C. $k = 14$

D. $\theta = 36^\circ$

(DSE-2013 PII Q30)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. The L.C.M. of a^2+4a+4, a^2-4 and a^3+8 is

A. $a+2$

B. $(a-2)(a+2)^2(a^2-2a+4)$

C. $(a-2)(a+2)^2(a^2+2a+4)$

D. $(a-2)(a+2)^4(a^2-2a+4)$

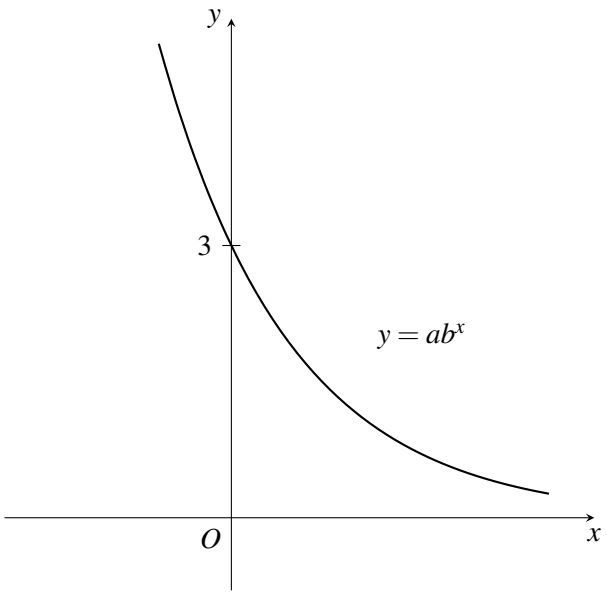
(DSE-2013 PII Q31)

Answers written in the margins will not be marked.

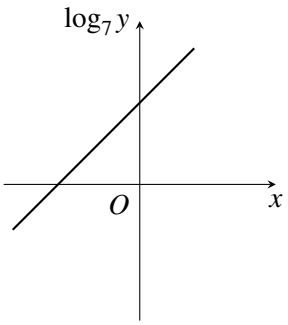
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

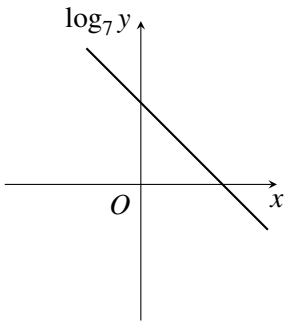
13. The figure below shows the graph of $y = ab^x$, where a and b are constants. Which of the following graphs may represent the relation between x and $\log_7 y$?



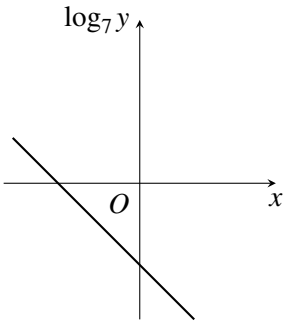
A.



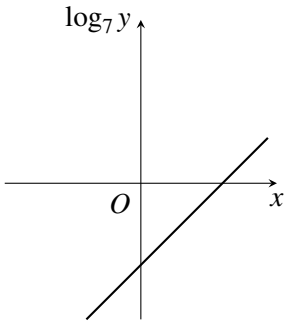
B.



C.



D.



(DSE-2013 PII Q32)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

14. If $x - \log y = x^2 - \log y^2 - 10 = 2$, then $y =$

A. 100

B. 2 or -4

C. $\frac{1}{100}$ or 10000

D. $\frac{1}{10000}$ or 100

(DSE-2013 PII Q34)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

15. The real part of $i+2i^2+3i^3+4i^4$ is

A. 2

B. -2

C. 6

D. -6

(DSE-2013 PII Q36)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

16. Consider the following system of inequalities:

$$\begin{cases} x \geq 2 \\ y \geq 0 \\ x + 4y \leq 22 \\ 4x - y \leq 20 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in D, then the greatest value of $3y - 4x + 15$ is

A. 3.

B. 17.

C. 22.

D. 30.

(DSE-2013 PII Q37)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

17. The nth term of a sequence is $2n - 19$. Which are true?

I. 25 is a term of the sequence..

II. The sequence has 10 negative terms.

III. The sum of the first n terms of the sequence is $n^2 - 18n$.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q38)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

18. If the height of a regular tetrahedron has height 2 cm. then the volume of the tetrahedron is

A. $2\,\text{cm}^3$

B. $\sqrt{3}\,\text{cm}^3$

C. $\sqrt{6}\,\text{cm}^3$

D. $3\sqrt{3}\,\text{cm}^3$

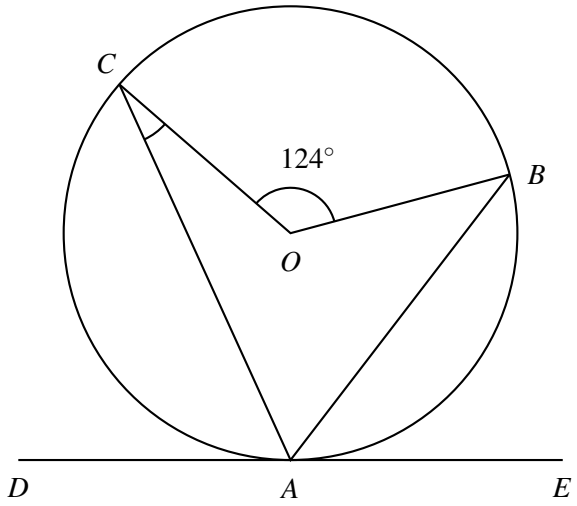
(DSE-2013 PII Q40)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

19. In the figure, O is the centre of the circle ABC, and DE is tangent to the circle at A. If AB is the angle bisector of $\angle CAE$, then $\angle ACO =$



A. 26°

B. 28°

C. 31°

D. 34°

(DSE-2013 PII Q41)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

20. Find the range of k such that the circle $x^2 + y^2 + 2x - 2y - 7 = 0$ and the straight line $3x - 4y + k = 0$ intersect.

A. $-8 < k < 22$

B. $-8 \leq k \leq 22$

C. $k < -22$ or $k > 8$

D. $k \leq -22$ or $k \geq 8$

(DSE-2013 PII Q42)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

21. If the variance of the five numbers x_1, x_2, x_3, x_4 and x_5 is 13, then the variance of the five numbers$3x_1 + 4, 3x_2 + 4, 3x_3 + 4, 3x_4 + 4$ and $3x_5 + 4$ is

A. 39

B. 43

C. 117

D. 121

(DSE-2013 PII Q45)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.